

DEPARTMENT OF COMPUTER SCIENCE

MASTER OF COMPUTER APPLICATION (MCA)

SEMESTER – III

ENTERPRISE JAVA TECHNOLOGIES

JDBC–ODBC PROGRAMMING ASSIGNMENT

Date: 18-08-2026

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JDBC–ODBC & Servlet - Practicals

Assignment - 1

- Q-1. Write a Java program to establish a connection between a Java application and a database using JDBC and display an appropriate message after successful connection.
- Q-2. Write a Java program using JDBC to create a Student table containing Student ID, Student Name, Course, Semester, and Marks.
- Q-3. Write a Java program using JDBC to insert student details into the Student table.
- Q-4. Write a Java program using JDBC to retrieve and display all records from the Student table using ResultSet.
- Q-5. Write a Java program using JDBC to search for a student record based on the Student ID entered by the user.
- Q-6. Write a Java program using JDBC to update the marks of a student based on the Student ID.
- Q-7. Write a Java program using JDBC to delete a student record based on the Student ID.
- Q-8. Write a Java program using PreparedStatement to insert student details into a database table.
- Q-9. Write a Java program using PreparedStatement to search for a student record using Student ID.
- Q-10. Write a menu-driven Java JDBC application to perform insertion, display, searching, updating, and deletion of student records.
- Q-11. Write a Java JDBC program to maintain employee details such as Employee ID, Name, Department, Designation, and Salary and display the records.
- Q-12. Write a Java JDBC program to update the salary of an employee based on Employee ID.
- Q-13. Write a Java JDBC program to maintain product details such as Product ID, Product Name, Category, Quantity, and Price.
- Q-14. Write a Java JDBC program using ResultSetMetaData to display the number of columns and column names of a database table.
- Q-15. Write a Java JDBC program using DatabaseMetaData to display information about the connected database and JDBC driver.
- Q-16. Write a Java JDBC program to demonstrate transaction management using commit() and rollback().
- Q-17. Write a Java JDBC program to insert multiple records into a database using batch processing.
- Q-18. Write a Java JDBC program to retrieve the maximum, minimum, average, and total salary from an Employee table using SQL aggregate functions.
- Q-19. Write a Java JDBC program to retrieve related information from two database tables using an SQL JOIN operation.
- Q-20. Write a Java JDBC program to calculate and display the total marks, percentage, and result of students using data retrieved from a database.
- Q-21. Write a Java JDBC program to validate a username and password using records stored in a database.
- Q-22. Write a Java JDBC program to search and display database records based on a name entered by the user.

Q-23. Write a Java JDBC program to maintain bank account details such as Account Number, Customer Name, Account Type, and Balance and perform database operations.

Q-24. Write a Java JDBC program to maintain library book details such as Book ID, Book Name, Author, Publisher, and Price and perform database operations.

Q-25. Develop a Java JDBC-based database application for managing student records with proper database connectivity, data insertion, retrieval, searching, updating, and deletion facilities.

Q-26. Write a Servlet program to accept student name and marks through an HTML form and display the total marks, percentage, grade, and result.

Q-27. Write a Servlet program to accept a number and determine whether it is even, odd, prime, or composite.

Q-28. Write a Servlet program to accept a string and display its reverse, number of characters, number of vowels, and whether it is a palindrome.

Q-29. Write a Servlet program to demonstrate the complete Servlet life cycle using `init()`, `service()`, and `destroy()` methods.

Q-30. Write a Servlet program to demonstrate `RequestDispatcher` using both `forward()` and `include()` methods.

Q-31. Write a Servlet program to demonstrate HTTP request and response handling by displaying request parameters, request method, URL, protocol, client IP address, and HTTP header information.

Q-32. Develop a Servlet-based login application using `HttpSession` that authenticates a user, maintains the user's session across multiple pages, displays the logged-in username, and provides a logout facility.

Q-33. Develop a Servlet-based application to demonstrate Cookie-based session tracking by creating, retrieving, updating, and deleting cookies.

Q-34. Develop a Servlet application that demonstrates multiple session-tracking techniques, including `HttpSession`, Cookies, and URL Rewriting, and displays the user information maintained by each technique.

Q-35. Develop a Servlet-based student result management application that accepts marks through an HTML form, validates the input, calculates total, percentage, grade, and result, and displays the formatted result.

Q-36. Write a Servlet program to establish a JDBC database connection and perform insertion and retrieval of student records.

Q-37. Develop a Servlet-based application to insert and display student records using HTML, Servlet, JDBC, and MySQL.

Q-38. Develop a Servlet-based application to search, update, and delete student records using Student ID and JDBC.

Q-39. Develop a Servlet-based CRUD application for managing student records using HTML, Servlet, JDBC, `PreparedStatement`, and MySQL.

Q-40. Develop a Servlet-based Employee Management System that allows the user to add, search, update, delete, and display employee records using Servlet, JDBC, and MySQL.

Q-41. Develop a Servlet-based Bank Account Management System that maintains account details and provides facilities for creating accounts, searching accounts, updating account information, and displaying account details using JDBC.

Q-42. Develop a Servlet-based application that performs login authentication using database records, maintains the authenticated user's session, restricts access to authorized pages, and provides logout functionality.

Q-43. Develop a Servlet-based application to implement a search facility with multiple criteria, allowing users to search student records by Student ID, name, course, or semester using JDBC and `PreparedStatement`.

Q-44. Develop a Servlet-based application to display student records with sorting and filtering options, such as sorting by name or marks and filtering students based on course or semester.

Q-45. Develop a Servlet-based application to generate a student marks report by retrieving records from MySQL and displaying total marks, percentage, grade, pass/fail status, and class-wise information.

Q-46. Develop a Servlet-based application using Servlet, JDBC, `HttpSession`, Cookies, and `RequestDispatcher` to implement a complete student management system with login, authentication, CRUD operations, session management, and logout.

Q-47. Develop a complete Web-Based Student Management System using HTML, Servlet, JDBC, MySQL, and session management. The application should provide user login, authentication, student registration, search, display, update, delete, result calculation, session-based access control, and logout facilities.

Q-48. Develop a Servlet-based Online Shopping Cart Application that allows a user to log in, view products retrieved from a database, add products to a cart, update quantities, remove products, calculate the total amount, and complete the checkout process using HttpSession and JDBC.

Q-49. Develop a Servlet-based Employee Payroll Management System that stores employee details in a database and calculates gross salary, deductions, and net salary based on employee information.

Q-50. Develop a Servlet-based Hospital Patient Management System that allows authorized users to register patients, search patient records, update patient information, and display patient details using Servlet, JDBC, and MySQL.

Q-51. Develop a Servlet-based Course Registration System that allows students to log in, view available courses, register for courses, cancel registration, and display their registered courses using Servlet, JDBC, and HttpSession.

Q-52. Develop a Servlet-based Online Feedback System that allows authenticated users to submit feedback, stores the feedback in a database, and provides an administrator facility to view and analyze submitted feedback.

Q-53. Develop a Servlet-based Student Attendance Management System that allows faculty to record attendance, update attendance records, search attendance by student or date, and generate an attendance report using Servlet and JDBC.

Q-54. Develop a Servlet-based Result Management System that allows an administrator to enter and update student marks, while students can log in and view their result, percentage, grade, and pass/fail status.

Q-55. Develop a complete College Management System using HTML, Servlet, JDBC, MySQL, HttpSession, Cookies, and RequestDispatcher. The system should provide authentication and separate facilities for managing students, courses, attendance, examination results, and logout.